Backend Architecture (Express + Firebase)
Mobile App Architecture (Ionic/Angular + Capacitor)

verifica token

Backend Architecture (Express + Firebase)

Backend — Express.js :3000

Clientes

Ionic + Angular localhost:8100

Capacitor iOS / Android

CORS Middleware

auth.middleware.js Verifica token · RBAC

Firebase (Google Cloud)

Firebase Auth ID Tokens

Firestore NoSQL Database

Controllers

auth.controller

students.controller

courses.controller

evaluation.controller

users.controller

dashboard.controller

Models / Validation

student.model

course.model

evaluation.model

user.model

Routes /api/

/auth

/students

/courses

/evaluations

/users

/dashboard

Mobile App Architecture (Ionic/Angular + Capacitor)

Ionic / Angular App (Capacitor)

Entry Point

AppModule / AppComponent

App Routing Module

Auth Module

LoginPage

RegisterPage

Core Services

AuthService BehaviorSubject currentUser$ JWT Token → localStorage

ApiService HTTP Client + Bearer Token

StudentService

State / Storage

localStorage auth_user · auth_token

RxJS Observables

Backend REST API (localhost:3000/api)

/auth/login /auth/register

/students (CRUD)

/courses (CRUD)

/evaluations (CRUD)

/users/teachers

/dashboard/metrics /dashboard/performance

Feature Modules (Lazy Loaded)

Admin Tabs

DashboardPage

TeachersListPage TeacherFormPage

StudentsPage

CoursesPage CourseStudentsPage

EvaluationListPage EvaluationFormPage

Teacher Tabs

DashboardPage

CoursesPage

StudentsPage

EvaluationListPage EvaluationFormPage

Student Tabs

DashboardPage

CoursesPage

EvaluationListPage

Core Guards

authGuard

roleGuard

Firebase

Firebase Auth signOut

Firestore DB

Capacitor Plugins

Camera

Push Notifications

Haptics

Status Bar